# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

## ASSESSMENT TOOL 3 – SHORT ANSWER TEST

| | | | |
|---|---|---|---|
| **Course Code:** | CSA0309 | **Course Title:** | Data Structures |
| **CO Assessed:** | CO5 – Manipulation (BL3, BL4) | **Assessment:** | Assessment Tool 3 – Short Answer Test |
| **Weightage:** | 30% of CIA | **Total Marks:** | 100 |
| **CO2 Statement:** | *Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.* | | |
| **Student Name:** | V. YESHWANTH | **Reg. No.:** | 192511300 |
| **Section / Batch:** | | **Date:** | 02/09/2026 |

**Code of Conduct**

*I* V·Yeshwanth (192511300) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: V.Yesh

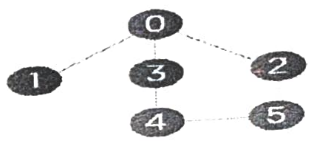
**Instructions**

- This test contains 10 short answer test questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

| Section / Topic | Focus | Q Nos | Marks |
|---|---|---|---|
| Minimum Spanning Tree | Kruskal's Algorithm | 1 | 20 |
| Graph Sorting | Topological Sort | 2 | 20 |
| Shortest Path Algorithm | MST & Dijkstra's Algorithm | 3 | 20 |
| Shortest Path Algorithm | Dijkstra's Algorithm | 4 | 20 |
| Minimum Spanning Tree | Prim's or Kruskal's Algorithm | 5 | 20 |
| | GRAND TOTAL: | 100 Marks | |

## Annexure A – Short Answer Test

1. Perform the following operations in the Singly Linked List for the following input.
   Input: 22-> 77-> 55-> 88->
   i. Insert a new-node 110 at the end of the list.
   ii. Insert a new-node 101 at the position 4 of the list.
   iii. Delete a node at the beginning of the single linked list
   iv. Delete a node at the position 3 of the single linked list.
2. Rotations in each step of insertion. 1,2,3,4,5,6,7,8,9,10 (), 4) Push (90), 5) Push (36), 6) Push (11), 7) Push (88), 8) Pop (), 9) Pop (). Draw the diagram of the stack and illustrate the above operations and identify where the top is?
3. Convert the following infix into a postfix expression. → B - (C/D + (E % F* G) / H) * J
4. Evaluate the given postfix expression. 53+62/*35*+
5. Construct a B-Tree of Order 3 by inserting numbers 34,26,45,12,87,56,78,22,23,24,65,85,95.
6. Construct a 2-3-4 Tree by inserting the following sequence of numbers 45, 36, 76, 23, 89, 115, 98, 39, 41, 56, 69, 48.
7. Build a max heap for the following numbers - 20, 35, 23, 22, 4, 45, 21, 5, 42 and 19. At each stage of heap tree construction, ensure heap properties are satisfied.
8. Perform the operations onto splay tree (draw after each operation): insert (3), insert (9), insert (12), insert (55), insert (1), insert (3), delete (3), delete (9), delete (60), insert (3), search (12).
9. Describe the algorithm to sort the elements 170, 45, 75, 90, 802, 24, 2, 66 using Radix sort. Explain with each phase?
10. Consider the following graph & construct BFS.




Graph with six nodes

**Rubrics for Calculation**

| Criteria | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---|---|---|---|---|---|
| Conceptual Understanding | 30 | Demonstrates complete and accurate understanding of concepts | Good understanding with minor gaps | Basic understanding | Limited understanding |
| Accuracy of Answer | 25 | Answers are completely correct and relevant | Mostly correct with minor errors | Partially correct | Mostly incorrect |
| Problem-Solving & Reasoning | 20 | Provides logical reasoning and appropriate steps | Good reasoning with minor gaps | Basic reasoning | Lacks logical reasoning |
| Clarity & Relevance | 15 | Answers are precise, clear, concise, and directly address the question | Mostly clear and relevant | Somewhat unclear or incomplete | Irrelevant or unclear answers |
| Time Management & CompletionS | 10 | Completes all questions accurately within the time | Completes most questions on time | Requires additional time | Unable to complete the test |

| SFaculty In-charge | Course Coordinator | Program Director |
|---|---|---|
| Signature: | Signature: ___________ | Signature: ___________ |
| Date: ___________ | Date: ___________ | Date: ___________ |

After push (88):

TOP
|
88
11
36
90.

After first pop () → 88 removed.

TOP
|
11
36
90

After second pop () → 11 removed.

TOP
|
36
90

Final TOP = 36

data structure : Stack : principle. LIFO

3.

INFIX to POSTFIX.

INFIX :

B - (C/(D + ((.%. F * G) / H)) * J

Conversion.

*, /, % → higher priority

+, - → lower priority

POSTFIX :

BCD) EF-/·G * H /IJ *

You can write it without spaces

B CD (EF-/·G* H/+J*_

4.

Postix :

53 + 62/* 35*+

use stract

| Token | operation | stack |
|---|---|---|
| 5 | push | 5 |
| 3 | push | 5,3 |
| + | 5 +3 | 8 |
| 6 | push | 8 ,6 |
| 2 | push | 8,6,2 |
| / | 6 /2 | 8,3 |
| + | 8 ×3 | 24 |
| 3 | push | 24,3 |
| 5 | push | 24, 3, 5 |
| d | 3×5 | 24, 15 |
| + | 24+15 | 39. |

Ans: 39

nal List.

77 → 55 → 88 → 110 → Null

ALGORITHM:

1. Traverse the linked list
2. For insertion, create a new mode.
3. Adjust the requried links.
4. For deletion, change the previous node's link.
5. Display the Final list.

5.

STACK OPERATION.

1. push (30)
2. push (36)
3. push (11)
4. push (88)
5. pop ( )
6. pop ( )

STACK CONSTRUCTION:

Initially:

Top
↑
90

After push (36).

Top
36.

Perform the following operations in the singly linked list for following input.

Input : 22 → 77 → 55 → 88 →

(i) insert a new node 110 at end of the list

(ii) insert a new-node 101 at the position 4 of the list.

(iii) Delete a node at the beginning of the singly linked list.

(iv) Delete a node at the position 3 of the singly linked list.

SINGLY linked LIST operation:

Input:

22 → 77 → 55 → 88 → Null.

(i) Insert 110 at the end.

22 → 77 → 55 → 88 → 110 → Null.

(ii) insert 101 at position 4

22 → 77 → 55 → 101 → 88 → 110 → Null

(iii) Delete node at begining.

Delete 22.

77 → 55 → 101 → 88 → 110 → Null

(iv). Delete node at position 3

position 3 = 101

45, 36, 76, 23, 89, 115, 98, 39, 41, 56, 69, 48

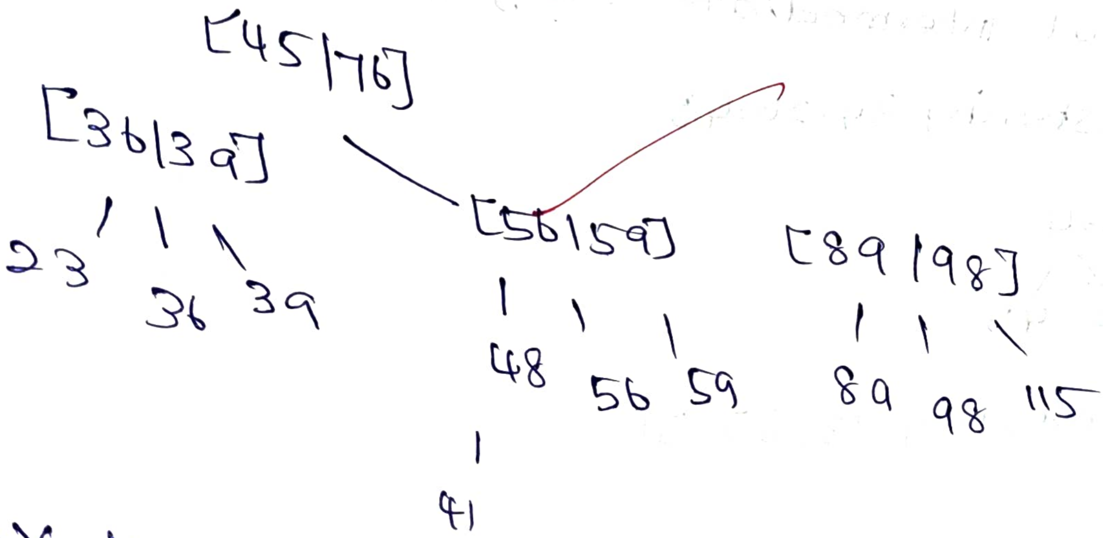
2-3-4 tree can contain

1 key → 2-node.

2 key → 3-node

3 keys → 4 -node

Find tree.



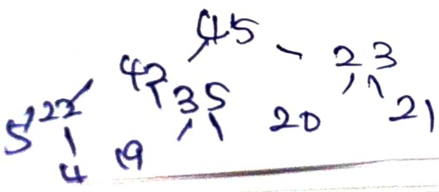

7. Max Heap.

Numbers:

20, 35, 23, 22, 4, 45, 21, 5, 42, 19.

In a max heap every parent must be greater than children.

Final Representation.



umber. 34, 26, 45, 12, 87, 56, 78, 22, 23, 24, 65, 85,
95, 85, 95

B-Tree of order 3.
Insert::

34, 26, 45, 12, 87, 56, 78, 22, 23, 24, 65, 85, 95
For an order 3 B tree - maximum children =
3 and maximum keys = 2
Importanat intermediate stages
After instering 34, 26, 45.

34
26 45

After inserting 12, 87, 56

34
12 26 45 87.

Continue splitting overlowing nodes
vaild find structure is

[34 | 56]

(22)(26) (45) (78 | 87)

12 23 24 45 56 65 85 95.

15, 40, 23, 22, 35, 20, 21, 5, 4, 19.

Notation:

Property.

For every node

Parent ≥ left child and right child.

Therefore, the above is a valid max Heap.

8,

Operations:

Insert [3], Insert(9). Insert(12), insert(50), Insert(3), Delete(3) Delete(9), Delete(60) Insert(8) Search(12)

Import Rule:

In a splay tree, after every insertion, deletion, a search the accessed node is moved to the root using rotations:

FIRST OPERATIONS:

After Insert (3):

3.

After Insert(9):

```
    9
   /
  3.
```

After Insert(55).:

```
  55
  /
 12
 /
9
/
3.
```

After Insert (1) the inserted node is splayed to the root.

```
1
 \
  55
  /
 12
 /
9
/
3.
```

For duliplication Inser(3), normally the existing node is accessed (spalyed rather than creating a duplicate Delete(60) → 60 doesnot exist

9. RADIX SORT.

element.

170, 45, 75, 90, 802, 24, 2, 66.

Radix sort processing digits right to left. using a stable sorting method.

Phase 1 - unit digit.

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| 90<br>170 | 7 | 2<br>802 | | 24 | 75<br>45 | 66 | | | | |

Phase 2 - Tendigits.

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| 2<br>802 | | 24 | | 45 | | 66 | 75<br>190 | | 90 | |

90
Phase 3 - Hundreds digits

| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| 66<br>95<br>24<br>2 | 802<br>170 | | | | | | | | | |

Final Answer :.

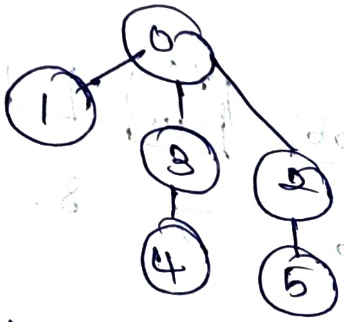
2, 24, 45, 66, 75, 90, 170, 802.

ALGORTHIM:

1. Find the maximum no. of. digits.
2. Sort number accordeng to unit digits, tendigits. hunderd digit.
3. continue unit 11 a digit position are processed.

4. The resulting sequence is sorted.

b. Consider the following graph & contract BFS




Problem understanding.

Breadth first search (BFS) is a graph traversal technique that visit nodes level. It uses a queue data structure to store and process vertices.

ALGORITHM.

1. Start from the given vertex 0 and mark it visited.

2. Insert vertex 0 into the queue

3. Remove a vertex from the queue and visit its unvisited adjacent vertices.

ALGORITHM:

1. Start from the given vertex 0 and make it visited.
2. Insert vertex 0 into the queue.
3. Remove a vertex from the queue and visit its unvisited adjacent vertices.
4. Add newly visited vertices to the queue.
5. Repeat until the queue becomes empty.

BFS traversal:

0 → 1 → 3 → 2 → 4 → 5

RESULT:

Thus, the graph is successfully traversed using BFS starting from vertex 0.